

NBA API Exploration Notebook

This notebook systematically explores every major endpoint of the `nba_api` library to catalog available features for building ML models to predict NBA win percentages.

Models planned:

1. **Team vs Team Model** — Capture playstyle matchups (pace, shooting profiles, defense schemes)
2. **Player Model** — Individual player impact that feeds into the team model
3. **Additional models** suggested at the end

Approach: For each endpoint, we call it with sample parameters, display `.head()` and `.dtypes`, then summarize useful features.

⚠ The NBA API rate-limits requests. We add `time.sleep()` between calls. Some endpoints may occasionally timeout.

```
In [1]: import pandas as pd
import time
from nba_api.stats.static import teams, players

pd.set_option('display.max_columns', None)
pd.set_option('display.max_colwidth', 40)

# Helper to safely call an endpoint and display info
def explore(name, df):
    """Display head and dtypes for a DataFrame returned by an nba_api endpoint"""
    print(f"\n{'='*80}")
    print(f"📊 {name}")
    print(f"    Shape: {df.shape} | Columns: {df.shape[1]}")
    print(f"{'='*80}")
    display(df.head(3))
    print(f"\n📋 Data Types:")
    display(df.dtypes)
    return df
```

1. Static Data (No API Call Required)

These are locally cached lookups — team IDs, player IDs, etc.

```
In [2]: # --- Static: All NBA Teams ---
nba_teams = teams.get_teams()
df_teams = pd.DataFrame(nba_teams)
explore("Static Teams", df_teams)

print("\n")
```

```
# --- Static: All NBA Players ---
nba_players = players.get_players()
df_players = pd.DataFrame(nba_players)
explore("Static Players", df_players)
```



Static Teams
Shape: (30, 7) | Columns: 7

	id	full_name	abbreviation	nickname	city	state	year_found
0	1610612737	Atlanta Hawks	ATL	Hawks	Atlanta	Georgia	19
1	1610612738	Boston Celtics	BOS	Celtics	Boston	Massachusetts	19
2	1610612739	Cleveland Cavaliers	CLE	Cavaliers	Cleveland	Ohio	19



Data Types:
id int64
full_name str
abbreviation str
nickname str
city str
state str
year_founded int64
dtype: object



Static Players
Shape: (5103, 5) | Columns: 5

	id	full_name	first_name	last_name	is_active
0	76001	Alaa Abdelnaby	Alaa	Abdelnaby	False
1	76002	Zaid Abdul-Aziz	Zaid	Abdul-Aziz	False
2	76003	Kareem Abdul-Jabbar	Kareem	Abdul-Jabbar	False



Data Types:
id int64
full_name str
first_name str
last_name str
is_active bool
dtype: object

Out [2]:

	id	full_name	first_name	last_name	is_active
0	76001	Alaa Abdelnaby	Alaa	Abdelnaby	False
1	76002	Zaid Abdul-Aziz	Zaid	Abdul-Aziz	False
2	76003	Kareem Abdul-Jabbar	Kareem	Abdul-Jabbar	False
3	51	Mahmoud Abdul-Rauf	Mahmoud	Abdul-Rauf	False
4	1505	Tariq Abdul-Wahad	Tariq	Abdul-Wahad	False
...	...	...	...	...	...
5098	78650	Matt Zunic	Matt	Zunic	False
5099	1641783	Tristan da Silva	Tristan	da Silva	True
5100	1628427	Vlatko Čančar	Vlatko	Čančar	False
5101	1642365	Nikola Đurišić	Nikola	Đurišić	True
5102	203967	Dario Šarić	Dario	Šarić	True

5103 rows x 5 columns

2. Team-Level Endpoints

Endpoints that return team-level stats — the core of the **Team vs Team model**.

We'll use the **Los Angeles Lakers (1610612747)** and season **2024-25** as sample parameters.

```
In [3]: from nba_api.stats.endpoints import (
    teamgamelog, teamgamelogs, teamestimatedmetrics,
    teaminfocommon, teamdashboardbygeneralsplits, teamdashboardbyshootingspl
    teamdashlineups, teamdashptpass, teamdashptreb, teamdashptshots,
    teamplayerdashboard, teamplayeronoffdetails, teamplayeronoffsummary,
    teamyearbyyearstats, teamdetails,
    leaguedashteamstats, leaguedashteamclutch, leaguedashteamptshot,
    leaguedashteamshotlocations, leaguedashptteamdefend,
    leaguehustlestatsteam, leaguestandingsv3
)

TEAM_ID = 1610612747 # Lakers
SEASON = '2024-25'

# Store all explored DataFrames for reference
explored = {}
```

```
In [4]: # 2a. TeamGameLog – per-game results for a team
    tgl = teamgamelog.TeamGameLog(team_id=TEAM_ID, season=SEASON)
    df = tgl.get_data_frames()[0]
    explored['TeamGameLog'] = explore("TeamGameLog – Per-game W/L, points, FG%,
```

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TeamGameLog – Per-game W/L, points, FG%, etc.
Shape: (82, 27) | Columns: 27
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```

	Team_ID	Game_ID	GAME_DATE	MATCHUP	WL	W	L	W_PCT	MIN	FGM
0	1610612747	0022401199	APR 13, 2025	LAL @ POR	L	50	32	0.610	240	31
1	1610612747	0022401185	APR 11, 2025	LAL vs. HOU	W	50	31	0.617	240	53
2	1610612747	0022401159	APR 09, 2025	LAL @ DAL	W	49	31	0.613	240	42

Data Types:

```
Team_ID      int64
Game_ID      str
GAME_DATE    str
MATCHUP      str
WL           str
W            int64
L            int64
W_PCT        float64
MIN          int64
FGM          int64
FGA          int64
FG_PCT       float64
FG3M         int64
FG3A         int64
FG3_PCT      float64
FTM          int64
FTA          int64
FT_PCT       float64
OREB         int64
DREB         int64
REB          int64
AST          int64
STL          int64
BLK          int64
TOV          int64
PF           int64
PTS          int64
dtype: object
```

```
In [5]: # 2b. TeamEstimatedMetrics – RPM-style estimated metrics per team
time.sleep(1)
tem = teamestimatedmetrics.TeamEstimatedMetrics(season=SEASON)
df = tem.get_data_frames()[0]
explored['TeamEstimatedMetrics'] = explore("TeamEstimatedMetrics – Estimated
```

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TeamEstimatedMetrics - Estimated OFF/DEF ratings, pace, W%
Shape: (30, 30) | Columns: 30
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```

	TEAM_NAME	TEAM_ID	GP	W	L	W_PCT	MIN	E_OFF_RATING	E_DEF_RATING
0	Memphis Grizzlies	1610612763	82	48	34	0.585	3941.0	114.4	109.1
1	Chicago Bulls	1610612741	82	39	43	0.476	3951.0	111.9	110.1
2	Atlanta Hawks	1610612737	82	40	42	0.488	3956.0	111.9	110.1

```
📦 Data Types:
TEAM_NAME          str
TEAM_ID            int64
GP                 int64
W                  int64
L                  int64
W_PCT              float64
MIN                float64
E_OFF_RATING       float64
E_DEF_RATING       float64
E_NET_RATING       float64
E_PACE             float64
E_AST_RATIO        float64
E_OREB_PCT         float64
E_DREB_PCT         float64
E_REB_PCT          float64
E_TM_TOV_PCT       float64
GP_RANK            int64
W_RANK             int64
L_RANK             int64
W_PCT_RANK         int64
MIN_RANK           int64
E_OFF_RATING_RANK  int64
E_DEF_RATING_RANK  int64
E_NET_RATING_RANK  int64
E_AST_RATIO_RANK   int64
E_OREB_PCT_RANK    int64
E_DREB_PCT_RANK    int64
E_REB_PCT_RANK     int64
E_TM_TOV_PCT_RANK  int64
E_PACE_RANK        int64
dtype: object
```

```
In [6]: # 2c. LeagueDashTeamStats - comprehensive team stats (Base, Advanced, Four Factors)
time.sleep(1)

# Base stats
ldts_base = leaguedashteamstats.LeagueDashTeamStats(season=SEASON, measure_type='Base')
df = ldts_base.get_data_frames()[0]
explored['LeagueDashTeamStats_Base'] = explore("LeagueDashTeamStats (Base) -
```

```
time.sleep(1)

# Advanced stats (OFF_RATING, DEF_RATING, NET_RATING, PACE, PIE, etc.)
ldts_adv = leaguedashteamstats.LeagueDashTeamStats(season=SEASON, measure_ty
df = ldts_adv.get_data_frames()[0]
explored['LeagueDashTeamStats_Advanced'] = explore("LeagueDashTeamStats (Adv

time.sleep(1)

# Four Factors
ldts_ff = leaguedashteamstats.LeagueDashTeamStats(season=SEASON, measure_ty
df = ldts_ff.get_data_frames()[0]
explored['LeagueDashTeamStats_FourFactors'] = explore("LeagueDashTeamStats (

time.sleep(1)

# Opponent stats
ldts_opp = leaguedashteamstats.LeagueDashTeamStats(season=SEASON, measure_ty
df = ldts_opp.get_data_frames()[0]
explored['LeagueDashTeamStats_Opponent'] = explore("LeagueDashTeamStats (Opp
```

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 LeagueDashTeamStats (Base) – Traditional team stats
Shape: (30, 54) | Columns: 54

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	TEAM_ID	TEAM_NAME	GP	W	L	W_PCT	MIN	FGM	FGA	FG_PCT	FG3M
0	1610612737	Atlanta Hawks	82	40	42	0.488	3956.0	3556	7528	0.472	1107
1	1610612738	Boston Celtics	82	61	21	0.744	3966.0	3413	7382	0.462	1457
2	1610612751	Brooklyn Nets	82	26	56	0.317	3956.0	3086	7060	0.437	1113

 Data Types:

TEAM_ID	int64
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64
PLUS_MINUS_RANK	int64
dtype: object	

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 LeagueDashTeamStats (Advanced) – OFF/DEF Rating, Pace, eFG%, TS%

Shape: (30, 46) | Columns: 46

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	TEAM_ID	TEAM_NAME	GP	W	L	W_PCT	MIN	E_OFF_RATING	OFF_RATING
0	1610612737	Atlanta Hawks	82	40	42	0.488	3956.0	111.9	113.7
1	1610612738	Boston Celtics	82	61	21	0.744	3966.0	117.5	119.5
2	1610612751	Brooklyn Nets	82	26	56	0.317	3956.0	105.7	108.1

 Data Types:

```
TEAM_ID          int64
TEAM_NAME        str
GP               int64
W                int64
L                int64
W_PCT            float64
MIN              float64
E_OFF_RATING     float64
OFF_RATING       float64
E_DEF_RATING     float64
DEF_RATING       float64
E_NET_RATING     float64
NET_RATING       float64
AST_PCT          float64
AST_TO           float64
AST_RATIO        float64
OREB_PCT         float64
DREB_PCT         float64
REB_PCT          float64
TM_TOV_PCT       float64
EFG_PCT          float64
TS_PCT           float64
E_PACE           float64
PACE             float64
PACE_PER40       float64
POSS             int64
PIE              float64
GP_RANK          int64
W_RANK           int64
L_RANK           int64
W_PCT_RANK       int64
MIN_RANK         int64
OFF_RATING_RANK  int64
DEF_RATING_RANK  int64
NET_RATING_RANK  int64
AST_PCT_RANK     int64
AST_TO_RANK      int64
AST_RATIO_RANK   int64
OREB_PCT_RANK    int64
DREB_PCT_RANK    int64
REB_PCT_RANK     int64
TM_TOV_PCT_RANK  int64
EFG_PCT_RANK     int64
TS_PCT_RANK      int64
PACE_RANK        int64
PIE_RANK         int64
dtype: object
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🇮🇹 LeagueDashTeamStats (Four Factors) – eFG%, TOV%, OREB%, FTA Rate
  Shape: (30, 28) | Columns: 28
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	TEAM_ID	TEAM_NAME	GP	W	L	W_PCT	MIN	EFG_PCT	FTA_RATE	TM_TOV
0	1610612737	Atlanta Hawks	82	40	42	0.488	3956.0	0.546	0.252	
1	1610612738	Boston Celtics	82	61	21	0.744	3966.0	0.561	0.212	
2	1610612751	Brooklyn Nets	82	26	56	0.317	3956.0	0.516	0.240	

Data Types:

TEAM_ID int64
TEAM_NAME str
GP int64
W int64
L int64
W_PCT float64
MIN float64
EFG_PCT float64
FTA_RATE float64
TM_TOV_PCT float64
OREB_PCT float64
OPP_EFG_PCT float64
OPP_FTA_RATE float64
OPP_TOV_PCT float64
OPP_OREB_PCT float64
GP_RANK int64
W_RANK int64
L_RANK int64
W_PCT_RANK int64
MIN_RANK int64
EFG_PCT_RANK int64
FTA_RATE_RANK int64
TM_TOV_PCT_RANK int64
OREB_PCT_RANK int64
OPP_EFG_PCT_RANK int64
OPP_FTA_RATE_RANK int64
OPP_TOV_PCT_RANK int64
OPP_OREB_PCT_RANK int64
dtype: object

LeagueDashTeamStats (Opponent) – What opponents shoot against this team
Shape: (30, 54) | Columns: 54

	TEAM_ID	TEAM_NAME	GP	W	L	W_PCT	MIN	OPP_FGM	OPP_FGA	OPP_F
0	1610612737	Atlanta Hawks	82	40	42	0.488	3956.0	3562.0	7398.0	
1	1610612738	Boston Celtics	82	61	21	0.744	3966.0	3302.0	7331.0	
2	1610612751	Brooklyn Nets	82	26	56	0.317	3956.0	3317.0	6909.0	

Data Types:

TEAM_ID	int64
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
OPP_FGM	float64
OPP_FGA	float64
OPP_FG_PCT	float64
OPP_FG3M	float64
OPP_FG3A	float64
OPP_FG3_PCT	float64
OPP_FTM	float64
OPP_FTA	float64
OPP_FT_PCT	float64
OPP_OREB	float64
OPP_DREB	float64
OPP_REB	float64
OPP_AST	float64
OPP_TOV	float64
OPP_STL	float64
OPP_BLK	float64
OPP_BLK_A	float64
OPP_PF	float64
OPP_PFD	float64
OPP_PTS	float64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
OPP_FGM_RANK	int64
OPP_FGA_RANK	int64
OPP_FG_PCT_RANK	int64
OPP_FG3M_RANK	int64
OPP_FG3A_RANK	int64
OPP_FG3_PCT_RANK	int64
OPP_FTM_RANK	int64
OPP_FTA_RANK	int64
OPP_FT_PCT_RANK	int64
OPP_OREB_RANK	int64
OPP_DREB_RANK	int64
OPP_REB_RANK	int64
OPP_AST_RANK	int64
OPP_TOV_RANK	int64
OPP_STL_RANK	int64
OPP_BLK_RANK	int64
OPP_BLK_A_RANK	int64
OPP_PF_RANK	int64
OPP_PFD1	int64
OPP_PTS_RANK	int64
PLUS_MINUS_RANK	int64
dtype:	object

```
In [7]: # 2d. LeagueDashTeamClutch – team performance in clutch situations
time.sleep(1)
ldtc = leaguedashteamclutch.LeagueDashTeamClutch(season=SEASON)
df = ldtc.get_data_frames()[0]
explored['LeagueDashTeamClutch'] = explore("LeagueDashTeamClutch – Clutch st
```

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 LeagueDashTeamClutch – Clutch stats (last 5 min, ±5 pts)
Shape: (30, 54) | Columns: 54

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	TEAM_ID	TEAM_NAME	GP	W	L	W_PCT	MIN	FGM	FGA	FG_PCT	FG
0	1610612737	Atlanta Hawks	41	21	20	0.512	160.035000	125	277	0.451	
1	1610612738	Boston Celtics	35	24	11	0.686	138.858333	118	246	0.480	
2	1610612751	Brooklyn Nets	38	17	21	0.447	145.363333	98	234	0.419	

 Data Types:

TEAM_ID	int64
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64
PLUS_MINUS_RANK	int64
dtype: object	

```
In [8]: # 2e. LeagueHustleStatsTeam – deflections, loose balls, contested shots, cha
time.sleep(1)
lhst = leaguehustlestatsteam.LeagueHustleStatsTeam(season=SEASON)
df = lhst.get_data_frames()[0]
explored['LeagueHustleStatsTeam'] = explore("LeagueHustleStatsTeam – Hustle
```

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LeagueHustleStatsTeam – Hustle stats (deflections, loose balls, etc.)
Shape: (30, 20) | Columns: 20
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```

	TEAM_ID	TEAM_NAME	MIN	CONTESTED_SHOTS	CONTESTED_SHOTS_2PT	C
0	1610612737	Atlanta Hawks	3927.0	3386	2015	
1	1610612738	Boston Celtics	3930.0	3555	2161	
2	1610612751	Brooklyn Nets	3923.0	3057	1741	

Data Types:

```
TEAM_ID                int64
TEAM_NAME              str
MIN                    float64
CONTESTED_SHOTS        int64
CONTESTED_SHOTS_2PT    int64
CONTESTED_SHOTS_3PT    int64
DEFLECTIONS            int64
CHARGES_DRAWN          int64
SCREEN_ASSISTS         int64
SCREEN_AST_PTS         int64
OFF_LOOSE_BALLS_RECOVERED int64
DEF_LOOSE_BALLS_RECOVERED int64
LOOSE_BALLS_RECOVERED  int64
PCT_LOOSE_BALLS_RECOVERED_OFF float64
PCT_LOOSE_BALLS_RECOVERED_DEF float64
OFF_BOXOUTS           int64
DEF_BOXOUTS           int64
BOX_OUTS              int64
PCT_BOX_OUTS_OFF      float64
PCT_BOX_OUTS_DEF      float64
dtype: object
```

```
In [9]: # 2f. LeagueStandingsV3 – conference/division standings, streaks, last 10
time.sleep(1)
ls = leaguestandingsv3.LeagueStandingsV3(season=SEASON)
df = ls.get_data_frames()[0]
explored['LeagueStandingsV3'] = explore("LeagueStandingsV3 – Full standings

time.sleep(1)

# 2g. LeagueDashTeamPtShot – team shooting by shot type (catch & shoot, pull
ldtps = leaguedashteamptshot.LeagueDashTeamPtShot(season=SEASON)
df = ldtps.get_data_frames()[0]
```

```
explored['LeagueDashTeamPtShot'] = explore("LeagueDashTeamPtShot – Catch & shoot, pull-up, paint touch shooting")
time.sleep(1)

# 2h. LeagueDashPtTeamDefend – opponent shot defense by zone/distance
ldptd = leaguedashptteamdefend.LeagueDashPtTeamDefend(season=SEASON, defense=DEFENSE)
df = ldptd.get_data_frames()[0]
explored['LeagueDashPtTeamDefend'] = explore("LeagueDashPtTeamDefend – Shot defense by zone/distance")
time.sleep(1)

# 2i. TeamDashboardByGeneralSplits – home/away, pre/post all-star, by month, by season
tdgs = teamdashboardbygeneralsplits.TeamDashboardByGeneralSplits(team_id=TEAM_ID)
dfs = tdgs.get_data_frames()
explored['TeamDashboardByGeneralSplits_Overall'] = explore("TeamDashboardByGeneralSplits_Overall")
explored['TeamDashboardByGeneralSplits_ByLocation'] = explore("TeamDashboardByGeneralSplits_ByLocation")
```

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LeagueStandingsV3 – Full standings (W/L, streak, L10, home/road)
Shape: (30, 92) | Columns: 92

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	LeagueID	SeasonID	TeamID	TeamCity	TeamName	TeamSlug	Conference	Con
0	00	22024	1610612739	Cleveland	Cavaliers	cavaliers	East	
1	00	22024	1610612760	Oklahoma City	Thunder	thunder	West	
2	00	22024	1610612738	Boston	Celtics	celtics	East	

Data Types:

```
LeagueID          str
SeasonID          str
TeamID            int64
TeamCity          str
TeamName          str
...
DiffTotalPoints   int64
LeagueGamesBack   float64
PlayoffSeeding    float64
ClinchedPostSeason int64
NEUTRAL           str
Length: 92, dtype: object
```

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LeagueDashTeamPtShot – Catch & shoot, pull-up, paint touch shooting
Shape: (30, 18) | Columns: 18

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	TEAM_ID	TEAM_NAME	TEAM_ABBREVIATION	GP	G	FGA_FREQUENCY	FGM
0	1610612745	Houston Rockets	HOU	82	82	1.0	3484
1	1610612760	Oklahoma City Thunder	OKC	82	82	1.0	3660
2	1610612763	Memphis Grizzlies	MEM	81	82	1.0	3625

Data Types:

TEAM_ID	int64
TEAM_NAME	str
TEAM_ABBREVIATION	str
GP	int64
G	int64
FGA_FREQUENCY	float64
FGM	int64
FGA	int64
FG_PCT	float64
EFG_PCT	float64
FG2A_FREQUENCY	float64
FG2M	int64
FG2A	int64
FG2_PCT	float64
FG3A_FREQUENCY	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64

dtype: object

LeagueDashPtTeamDefend – Shot defense (contested %, DFG%)
Shape: (30, 11) | Columns: 11

	TEAM_ID	TEAM_NAME	TEAM_ABBREVIATION	GP	G	FREQ	D_FGM	D_FGA	D_
0	1610612760	Oklahoma City Thunder	OKC	82	82	1.0	3105	7125	
1	1610612738	Boston Celtics	BOS	82	82	1.0	3299	7327	
2	1610612739	Cleveland Cavaliers	CLE	82	82	1.0	3382	7453	

Data Types:

TEAM_ID int64
TEAM_NAME str
TEAM_ABBREVIATION str
GP int64
G int64
FREQ float64
D_FGM int64
D_FGA int64
D_FG_PCT float64
NORMAL_FG_PCT float64
PCT_PLUSMINUS float64
dtype: object

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 TeamDashboardByGeneralSplits – Overall
Shape: (1, 55) | Columns: 55

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	GROUP_SET	GROUP_VALUE	SEASON_YEAR	GP	W	L	W_PCT	MIN	FGM	FG
0	Overall	2024-25	2024-25	82	50	32	0.61	3946.0	3356	700

 Data Types:

GROUP_SET	str
GROUP_VALUE	str
SEASON_YEAR	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64
PLUS_MINUS_RANK	int64
dtype: object	

=====

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 TeamDashboardByGeneralSplits – Home/Away Split
Shape: (2, 55) | Columns: 55

=====

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	GROUP_SET	GROUP_VALUE	TEAM_GAME_LOCATION	GP	W	L	W_PCT	MIN	F
0	Location	Home	Home	41	31	10	0.756	1973.0	1
1	Location	Road	Road	41	19	22	0.463	1973.0	1

 Data Types:

GROUP_SET	str
GROUP_VALUE	str
TEAM_GAME_LOCATION	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64
PLUS_MINUS_RANK	int64
dtype: object	

```
In [10]: # 2j. SynergyPlayTypes – playstyle breakdown (Isolation, PnR, Transition, et
from nba_api.stats.endpoints import synergyplaytypes, leaguedashptstats, lea

time.sleep(1)
spt = synergyplaytypes.SynergyPlayTypes(season=SEASON, player_or_team_abbrev
df = spt.get_data_frames()[0]
explored['SynergyPlayTypes_Team_Off'] = explore("SynergyPlayTypes (Team, Off

time.sleep(1)

# 2k. LeagueDashPtStats – tracking stats (speed, distance, touches, passes,
ldpts = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='T
df = ldpts.get_data_frames()[0]
explored['LeagueDashPtStats_SpeedDistance'] = explore("LeagueDashPtStats (Sp

time.sleep(1)

ldpts2 = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='
df = ldpts2.get_data_frames()[0]
explored['LeagueDashPtStats_Passing'] = explore("LeagueDashPtStats (Passing)

time.sleep(1)

ldpts3 = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='
df = ldpts3.get_data_frames()[0]
explored['LeagueDashPtStats_Possessions'] = explore("LeagueDashPtStats (Poss

time.sleep(1)

ldpts4 = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='
df = ldpts4.get_data_frames()[0]
explored['LeagueDashPtStats_Rebounding'] = explore("LeagueDashPtStats (Rebou
```

```
=====
=====
📊 SynergyPlayTypes (Team, Offense) – Iso, PnR, Transition, Post-Up, Spot-U
p, etc.
   Shape: (0, 22) | Columns: 22
=====
=====
```

SEASON_ID	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	PLAY_TYPE	TYPE_GROUP
-----------	---------	-------------------	-----------	-----------	------------

📄 Data Types:

SEASON_ID object
TEAM_ID object
TEAM_ABBREVIATION object
TEAM_NAME object
PLAY_TYPE object
TYPE_GROUPING object
PERCENTILE object
GP object
POSS_PCT object
PPP object
FG_PCT object
FT_POSS_PCT object
TOV_POSS_PCT object
SF_POSS_PCT object
PLUSONE_POSS_PCT object
SCORE_POSS_PCT object
EFG_PCT object
POSS object
PTS object
FGM object
FGA object
FGMX object
dtype: object

LeagueDashPtStats (Speed & Distance) – Avg speed, distance covered
Shape: (30, 14) | Columns: 14

	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	W	L	MIN	DIST_FEET	I
0	1610612737	ATL	Atlanta Hawks	82	40	42	19780.0	8127584	
1	1610612738	BOS	Boston Celtics	82	61	21	19830.0	7467732	
2	1610612751	BKN	Brooklyn Nets	82	26	56	19780.0	8088134	

Data Types:
TEAM_ID int64
TEAM_ABBREVIATION str
TEAM_NAME str
GP int64
W int64
L int64
MIN float64
DIST_FEET int64
DIST_MILES float64
DIST_MILES_OFF float64
DIST_MILES_DEF float64
AVG_SPEED float64
AVG_SPEED_OFF float64
AVG_SPEED_DEF float64
dtype: object

```
=====
=====
LeagueDashPtStats (Passing) – Passes made/received, assist opportunities
Shape: (30, 17) | Columns: 17
=====
=====
```

	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	W	L	MIN	PASSES_MADE
0	1610612737	ATL	Atlanta Hawks	82	40	42	19780.0	231
1	1610612738	BOS	Boston Celtics	82	61	21	19830.0	2252
2	1610612751	BKN	Brooklyn Nets	82	26	56	19780.0	2506

```
📋 Data Types:
TEAM_ID                int64
TEAM_ABBREVIATION      str
TEAM_NAME              str
GP                    int64
W                    int64
L                    int64
MIN                  float64
PASSES_MADE           int64
PASSES_RECEIVED       int64
AST                  int64
FT_AST               int64
SECONDARY_AST         int64
POTENTIAL_AST         int64
AST_PTS_CREATED       int64
AST_ADJ              int64
AST_TO_PASS_PCT       float64
AST_TO_PASS_PCT_ADJ   float64
dtype: object
```

```
=====
=====
LeagueDashPtStats (Possessions) – Touches, time of possession, dribbles
Shape: (30, 20) | Columns: 20
=====
=====
```

	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	W	L	MIN	POINTS	TOL
0	1610612737	ATL	Atlanta Hawks	82	40	42	19780.0	9691	
1	1610612738	BOS	Boston Celtics	82	61	21	19830.0	9534	
2	1610612751	BKN	Brooklyn Nets	82	26	56	19780.0	8619	

```
📋 Data Types:
```

TEAM_ID int64
TEAM_ABBREVIATION str
TEAM_NAME str
GP int64
W int64
L int64
MIN float64
POINTS int64
TOUCHES int64
FRONT_CT_TOUCHES int64
TIME_OF_POSS float64
AVG_SEC_PER_TOUCH float64
AVG_DRIB_PER_TOUCH float64
PTS_PER_TOUCH float64
ELBOW_TOUCHES int64
POST_TOUCHES int64
PAINT_TOUCHES int64
PTS_PER_ELBOW_TOUCH float64
PTS_PER_POST_TOUCH float64
PTS_PER_PAINT_TOUCH float64
dtype: object

=====

====

 LeagueDashPtStats (Rebounding) – Reb chances, contested/uncontested
Shape: (30, 34) | Columns: 34

=====

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	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	W	L	MIN	OREB	OREB.
0	1610612737	ATL	Atlanta Hawks	82	40	42	18926.0	974	
1	1610612738	BOS	Boston Celtics	82	61	21	19118.0	922	
2	1610612751	BKN	Brooklyn Nets	82	26	56	19171.0	894	

 Data Types:

```

TEAM_ID                int64
TEAM_ABBREVIATION      str
TEAM_NAME              str
GP                    int64
W                    int64
L                    int64
MIN                  float64
OREB                 int64
OREB_CONTEST         int64
OREB_UNCONTEST       int64
OREB_CONTEST_PCT     float64
OREB_CHANCES        int64
OREB_CHANCE_PCT     float64
OREB_CHANCE_DEFER    int64
OREB_CHANCE_PCT_ADJ  float64
AVG_OREB_DIST        float64
DREB                 int64
DREB_CONTEST         int64
DREB_UNCONTEST       int64
DREB_CONTEST_PCT     float64
DREB_CHANCES        int64
DREB_CHANCE_PCT     float64
DREB_CHANCE_DEFER    int64
DREB_CHANCE_PCT_ADJ  float64
AVG_DREB_DIST        float64
REB                  int64
REB_CONTEST          int64
REB_UNCONTEST        int64
REB_CONTEST_PCT      float64
REB_CHANCES          int64
REB_CHANCE_PCT       float64
REB_CHANCE_DEFER     int64
REB_CHANCE_PCT_ADJ   float64
AVG_REB_DIST         float64
dtype: object

```

3. Player-Level Endpoints

Endpoints that return individual player stats — the core of the **Player Model**.

We'll use **LeBron James (2544)** as a sample player.

```

In [11]: from nba_api.stats.endpoints import (
    playergamealog, playergamealogs, playercareerstats, playerestimatedmetrics,
    commonplayerinfo, leaguedashplayerstats, leaguedashplayerclutch,
    leaguedashplayerbiostats, leaguehustlestatsplayer,
    leaguedashplayerptshot, leaguedashptdefend,
    playerdashboardbygeneralsplits, playerdashptpass,
    playerdashptreb, playerdashptshots, playerdashptshotdefend,
    playerprofilev2, commonteamroster
)

PLAYER_ID = 2544 # LeBron James

```

```
In [12]: # 3a. PlayerGameLog – per-game stats for a single player
time.sleep(1)
pgl = playergamelog.PlayerGameLog(player_id=PLAYER_ID, season=SEASON)
df = pgl.get_data_frames()[0]
explored['PlayerGameLog'] = explore("PlayerGameLog – Per-game player stats",

time.sleep(1)

# 3b. CommonPlayerInfo – biographical data (height, weight, position, draft
cpi = commonplayerinfo.CommonPlayerInfo(player_id=PLAYER_ID)
df = cpi.get_data_frames()[0]
explored['CommonPlayerInfo'] = explore("CommonPlayerInfo – Bio/position/draf

time.sleep(1)

# 3c. PlayerCareerStats – season-by-season career totals
pcs = playercareerstats.PlayerCareerStats(player_id=PLAYER_ID)
dfs_career = pcs.get_data_frames()
explored['PlayerCareerStats_RegSeason'] = explore("PlayerCareerStats – Regul
```

```
=====
====
📊 PlayerGameLog – Per-game player stats
  Shape: (70, 27) | Columns: 27
=====
=====
```

	SEASON_ID	Player_ID	Game_ID	GAME_DATE	MATCHUP	WL	MIN	FGM	FGA
0	22024	2544	0022401185	Apr 11, 2025	LAL vs. HOU	W	22	6	11
1	22024	2544	0022401159	Apr 09, 2025	LAL @ DAL	W	36	11	20
2	22024	2544	0022401153	Apr 08, 2025	LAL @ OKC	L	35	8	19

📄 Data Types:

SEASON_ID str
Player_ID int64
Game_ID str
GAME_DATE str
MATCHUP str
WL str
MIN int64
FGM int64
FGA int64
FG_PCT float64
FG3M int64
FG3A int64
FG3_PCT float64
FTM int64
FTA int64
FT_PCT float64
OREB int64
DREB int64
REB int64
AST int64
STL int64
BLK int64
TOV int64
PF int64
PTS int64
PLUS_MINUS int64
VIDEO_AVAILABLE int64
dtype: object

=====
=====
 CommonPlayerInfo – Bio/position/draft info
Shape: (1, 33) | Columns: 33
=====
=====

	PERSON_ID	FIRST_NAME	LAST_NAME	DISPLAY_FIRST_LAST	DISPLAY_LAST_COMM
0	2544	LeBron	James	LeBron James	Jame

 Data Types:

PERSON_ID int64
FIRST_NAME str
LAST_NAME str
DISPLAY_FIRST_LAST str
DISPLAY_LAST_COMMA_FIRST str
DISPLAY_FI_LAST str
PLAYER_SLUG str
BIRTHDATE str
SCHOOL str
COUNTRY str
LAST_AFFILIATION str
HEIGHT str
WEIGHT str
SEASON_EXP int64
JERSEY str
POSITION str
ROSTERSTATUS str
GAMES_PLAYED_CURRENT_SEASON_FLAG str
TEAM_ID int64
TEAM_NAME str
TEAM_ABBREVIATION str
TEAM_CODE str
TEAM_CITY str
PLAYERCODE str
FROM_YEAR int64
TO_YEAR int64
DLEAGUE_FLAG str
NBA_FLAG str
GAMES_PLAYED_FLAG str
DRAFT_YEAR str
DRAFT_ROUND str
DRAFT_NUMBER str
GREATEST_75_FLAG str
dtype: object

PlayerCareerStats – Regular Season Totals
Shape: (23, 27) | Columns: 27

	PLAYER_ID	SEASON_ID	LEAGUE_ID	TEAM_ID	TEAM_ABBREVIATION	PLAYER_AG
0	2544	2003-04	00	1610612739	CLE	19
1	2544	2004-05	00	1610612739	CLE	20
2	2544	2005-06	00	1610612739	CLE	21

Data Types:

PLAYER_ID int64
SEASON_ID str
LEAGUE_ID str
TEAM_ID int64
TEAM_ABBREVIATION str
PLAYER_AGE float64
GP int64
GS int64
MIN int64
FGM int64
FGA int64
FG_PCT float64
FG3M int64
FG3A int64
FG3_PCT float64
FTM int64
FTA int64
FT_PCT float64
OREB int64
DREB int64
REB int64
AST int64
STL int64
BLK int64
TOV int64
PF int64
PTS int64
dtype: object

```
In [13]: # 3d. LeagueDashPlayerStats - league-wide player stats (Base + Advanced)
time.sleep(1)
ldps_base = leaguedashplayerstats.LeagueDashPlayerStats(season=SEASON, measu
df = ldps_base.get_data_frames()[0]
explored['LeagueDashPlayerStats_Base'] = explore("LeagueDashPlayerStats (Bas

time.sleep(1)

ldps_adv = leaguedashplayerstats.LeagueDashPlayerStats(season=SEASON, measur
df = ldps_adv.get_data_frames()[0]
explored['LeagueDashPlayerStats_Advanced'] = explore("LeagueDashPlayerStats
```

=====

=====
 LeagueDashPlayerStats (Base) - All players traditional stats
Shape: (569, 67) | Columns: 67

=====

=====

	PLAYER_ID	PLAYER_NAME	NICKNAME	TEAM_ID	TEAM_ABBREVIATION	AGE	GP
0	1630639	A.J. Lawson	A.J.	1610612761	TOR	24.0	26
1	1631260	AJ Green	AJ	1610612749	MIL	25.0	73
2	1642358	AJ Johnson	AJ	1610612764	WAS	20.0	29

 Data Types:

PLAYER_ID int64
PLAYER_NAME str
NICKNAME str
TEAM_ID int64
TEAM_ABBREVIATION str
...
NBA_FANTASY_PTS_RANK int64
DD2_RANK int64
TD3_RANK int64
WNBA_FANTASY_PTS_RANK int64
TEAM_COUNT int64
Length: 67, dtype: object

=====

====

 LeagueDashPlayerStats (Advanced) – OFF/DEF rating, USG%, TS%, PIE
Shape: (569, 79) | Columns: 79

=====

=====

	PLAYER_ID	PLAYER_NAME	NICKNAME	TEAM_ID	TEAM_ABBREVIATION	AGE	GP
0	1630639	A.J. Lawson	A.J.	1610612761	TOR	24.0	26
1	1631260	AJ Green	AJ	1610612749	MIL	25.0	73
2	1642358	AJ Johnson	AJ	1610612764	WAS	20.0	29

 Data Types:

PLAYER_ID int64
PLAYER_NAME str
NICKNAME str
TEAM_ID int64
TEAM_ABBREVIATION str
...
FGA_RANK int64
FGM_PG_RANK int64
FGA_PG_RANK int64
FG_PCT_RANK int64
TEAM_COUNT int64
Length: 79, dtype: object

```
In [14]: # 3e. PlayerEstimatedMetrics – estimated advanced stats per player (E_OFF_RA
time.sleep(1)
pem = playerestimatedmetrics.PlayerEstimatedMetrics(season=SEASON)
df = pem.get_data_frames()[0]
explored['PlayerEstimatedMetrics'] = explore("PlayerEstimatedMetrics – Estim
time.sleep(1)

# 3f. LeagueDashPlayerBioStats – height, weight, age distribution league-wic
ldpbs = leaguedashplayerbiostats.LeagueDashPlayerBioStats(season=SEASON)
df = ldpbs.get_data_frames()[0]
explored['LeagueDashPlayerBioStats'] = explore("LeagueDashPlayerBioStats – H
time.sleep(1)

# 3g. LeagueHustleStatsPlayer – individual hustle metrics
```

```
lhsp = leaguehustlestatsplayer.LeagueHustleStatsPlayer(season=SEASON)
df = lhsp.get_data_frames()[0]
explored['LeagueHustleStatsPlayer'] = explore("LeagueHustleStatsPlayer - Pla

time.sleep(1)

# 3h. LeagueDashPlayerClutch - player clutch performance
ldpc = leaguedashplayerclutch.LeagueDashPlayerClutch(season=SEASON)
df = ldpc.get_data_frames()[0]
explored['LeagueDashPlayerClutch'] = explore("LeagueDashPlayerClutch - Playe
```

=====

=====
 PlayerEstimatedMetrics - Estimated OFF/DEF rating, pace, W%
Shape: (569, 32) | Columns: 32

=====

	PLAYER_ID	PLAYER_NAME	GP	W	L	W_PCT	MIN	E_OFF_RATING	E_DEF_RATING
0	1627736	Malik Beasley	82	44	38	0.537	27.8	112.1	109.1
1	1642267	Bub Carrington	82	18	64	0.220	30.0	102.5	117.1
2	1630577	Julian Champagne	82	34	48	0.415	23.6	112.0	113.1

 Data Types:

PLAYER_ID int64
PLAYER_NAME str
GP int64
W int64
L int64
W_PCT float64
MIN float64
E_OFF_RATING float64
E_DEF_RATING float64
E_NET_RATING float64
E_AST_RATIO float64
E_OREB_PCT float64
E_DREB_PCT float64
E_REB_PCT float64
E_TOV_PCT float64
E_USG_PCT float64
E_PACE float64
GP_RANK int64
W_RANK int64
L_RANK int64
W_PCT_RANK int64
MIN_RANK int64
E_OFF_RATING_RANK int64
E_DEF_RATING_RANK int64
E_NET_RATING_RANK int64
E_AST_RATIO_RANK int64
E_OREB_PCT_RANK int64
E_DREB_PCT_RANK int64
E_REB_PCT_RANK int64
E_TOV_PCT_RANK int64
E_USG_PCT_RANK int64
E_PACE_RANK int64
dtype: object

=====

====

 LeagueDashPlayerBioStats – Height/Weight/Age/Experience per player
Shape: (569, 23) | Columns: 23

=====

=====

	PLAYER_ID	PLAYER_NAME	TEAM_ID	TEAM_ABBREVIATION	AGE	PLAYER_HEIGHT
0	1630639	A.J. Lawson	1610612761	TOR	24.0	6-
1	1631260	AJ Green	1610612749	MIL	25.0	6-
2	1642358	AJ Johnson	1610612764	WAS	20.0	6-

 Data Types:

PLAYER_ID int64
PLAYER_NAME str
TEAM_ID int64
TEAM_ABBREVIATION str
AGE float64
PLAYER_HEIGHT str
PLAYER_HEIGHT_INCHES int64
PLAYER_WEIGHT str
COLLEGE str
COUNTRY str
DRAFT_YEAR str
DRAFT_ROUND str
DRAFT_NUMBER str
GP int64
PTS int64
REB int64
AST int64
NET_RATING float64
OREB_PCT float64
DREB_PCT float64
USG_PCT float64
TS_PCT float64
AST_PCT float64
dtype: object

=====

====

 LeagueHustleStatsPlayer – Player hustle (deflections, loose balls, etc.)
Shape: (567, 28) | Columns: 28

=====

=====

	PLAYER_ID	PLAYER_NAME	TEAM_ID	TEAM_ABBREVIATION	AGE	G	MIN	CO
0	1630639	A.J. Lawson	1610612761	TOR	24.0	24	484.0	
1	1631260	AJ Green	1610612749	MIL	25.0	72	1656.0	
2	1642358	AJ Johnson	1610612764	WAS	20.0	22	616.0	

 Data Types:

PLAYER_ID int64
PLAYER_NAME str
TEAM_ID int64
TEAM_ABBREVIATION str
AGE float64
G int64
MIN float64
CONTESTED_SHOTS int64
CONTESTED_SHOTS_2PT int64
CONTESTED_SHOTS_3PT int64
DEFLECTIONS int64
CHARGES_DRAWN int64
SCREEN_ASSISTS int64
SCREEN_ASTPTS int64
OFF_LOOSE_BALLS_RECOVERED int64
DEF_LOOSE_BALLS_RECOVERED int64
LOOSE_BALLS_RECOVERED int64
PCT_LOOSE_BALLS_RECOVERED_OFF float64
PCT_LOOSE_BALLS_RECOVERED_DEF float64
OFF_BOXOUTS int64
DEF_BOXOUTS int64
BOX_OUT_PLAYER_TEAM_REBS int64
BOX_OUT_PLAYER_REBS int64
BOX_OUTS int64
PCT_BOX_OUTS_OFF float64
PCT_BOX_OUTS_DEF float64
PCT_BOX_OUTS_TEAM_REB float64
PCT_BOX_OUTS_REB float64
dtype: object

LeagueDashPlayerClutch – Player clutch stats
Shape: (498, 68) | Columns: 68

	GROUP_SET	PLAYER_ID	PLAYER_NAME	NICKNAME	TEAM_ID	TEAM_ABBREVIAT
0	Players	1630639	A.J. Lawson	A.J.	1610612761	-
1	Players	1631260	AJ Green	AJ	1610612749	
2	Players	1642358	AJ Johnson	AJ	1610612764	v

Data Types:

GROUP_SET str
PLAYER_ID int64
PLAYER_NAME str
NICKNAME str
TEAM_ID int64
...
NBA_FANTASY_PTS_RANK int64
DD2_RANK int64
TD3_RANK int64
WNBA_FANTASY_PTS_RANK int64
TEAM_COUNT int64
Length: 68, dtype: object

```
In [15]: # 3i. Player tracking stats (passing, rebounding, shooting)
time.sleep(1)
# Player Passing Tracking
ldpts_pp = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='Player')
df = ldpts_pp.get_data_frames()[0]
explored['LeagueDashPtStats_Player_Passing'] = explore("LeagueDashPtStats (Player, Passing) - Passes, potential assists")

time.sleep(1)
# Player Possessions Tracking
ldpts_pos = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='Player')
df = ldpts_pos.get_data_frames()[0]
explored['LeagueDashPtStats_Player_Possessions'] = explore("LeagueDashPtStats (Player, Possessions) - Possessions, potential assists")

time.sleep(1)
# Player Speed & Distance
ldpts_sd = leaguedashptstats.LeagueDashPtStats(season=SEASON, player_or_team='Player')
df = ldpts_sd.get_data_frames()[0]
explored['LeagueDashPtStats_Player_SpeedDist'] = explore("LeagueDashPtStats (Player, Speed & Distance) - Speed, Distance, Potential Assists")
```

=====

=====
 LeagueDashPtStats (Player, Passing) – Passes, potential assists
Shape: (569, 18) | Columns: 18

=====

=====

	PLAYER_ID	PLAYER_NAME	TEAM_ID	TEAM_ABBREVIATION	GP	W	L	MIN
0	1630639	A.J. Lawson	1610612761	TOR	26	14	12	486.0
1	1631260	AJ Green	1610612749	MIL	73	44	29	1659.0
2	1642358	AJ Johnson	1610612764	WAS	29	8	21	638.0

 Data Types:

PLAYER_ID	int64
PLAYER_NAME	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
GP	int64
W	int64
L	int64
MIN	float64
PASSES_MADE	int64
PASSES_RECEIVED	int64
AST	int64
FT_AST	int64
SECONDARY_AST	int64
POTENTIAL_AST	int64
AST_PTS_CREATED	int64
AST_ADJ	int64
AST_TO_PASS_PCT	float64
AST_TO_PASS_PCT_ADJ	float64

dtype: object

```
=====
=====
🏆 LeagueDashPtStats (Player, Possessions) – Touches, time of poss, dribbles
Shape: (569, 21) | Columns: 21
=====
=====
```

	PLAYER_ID	PLAYER_NAME	TEAM_ID	TEAM_ABBREVIATION	GP	W	L	MIN
0	1630639	A.J. Lawson	1610612761	TOR	26	14	12	486.0
1	1631260	AJ Green	1610612749	MIL	73	44	29	1659.0
2	1642358	AJ Johnson	1610612764	WAS	29	8	21	638.0

```
📄 Data Types:
PLAYER_ID          int64
PLAYER_NAME        str
TEAM_ID            int64
TEAM_ABBREVIATION  str
GP                 int64
W                  int64
L                  int64
MIN                float64
POINTS             int64
TOUCHES            int64
FRONT_CT_TOUCHES   int64
TIME_OF_POSS       float64
AVG_SEC_PER_TOUCH  float64
AVG_DRIB_PER_TOUCH float64
PTS_PER_TOUCH      float64
ELBOW_TOUCHES      int64
POST_TOUCHES       int64
PAINT_TOUCHES      int64
PTS_PER_ELBOW_TOUCH float64
PTS_PER_POST_TOUCH float64
PTS_PER_PAINT_TOUCH float64
dtype: object
=====
=====
```

```
=====
=====
🏆 LeagueDashPtStats (Player, SpeedDistance) – Avg speed, miles covered
Shape: (569, 15) | Columns: 15
=====
=====
```

	PLAYER_ID	PLAYER_NAME	TEAM_ID	TEAM_ABBREVIATION	GP	W	L	MIN
0	1630639	A.J. Lawson	1610612761	TOR	26	14	12	486.0
1	1631260	AJ Green	1610612749	MIL	73	44	29	1659.0
2	1642358	AJ Johnson	1610612764	WAS	29	8	21	638.0

```
📄 Data Types:
```

```

PLAYER_ID          int64
PLAYER_NAME         str
TEAM_ID            int64
TEAM_ABBREVIATION   str
GP                 int64
W                  int64
L                  int64
MIN                float64
DIST_FEET           int64
DIST_MILES          float64
DIST_MILES_OFF      float64
DIST_MILES_DEF      float64
AVG_SPEED           float64
AVG_SPEED_OFF       float64
AVG_SPEED_DEF       float64
dtype: object

```

4. Game-Level & Box Score Endpoints

Per-game data that forms the training labels and provides detailed box scores.

We'll use a sample Game ID from the 2024-25 season.

```

In [16]: from nba_api.stats.endpoints import (
    leaguegamelog, leaguegamefinder,
    boxscoretraditionalv3, boxscoreadvancedv3, boxscorehustlev2,
    boxscorefourfactorsv3, boxscoremiscv3, boxscoreplayertrackv3,
    boxscorescoringv3, boxscoreusagev3, boxscoresummaryv2,
    boxscorematchupsv3,
    winprobabilitypbp, playbyplayv3,
    commonteamroster, teamplayeronoffdetails,
    leaguedashlineups, leagueseasonmatchups
)

# Get a sample game ID from the Lakers' game log
GAME_ID = explored['TeamGameLog'].iloc[0]['Game_ID']
print(f"Sample Game ID: {GAME_ID}")

```

Sample Game ID: 0022401199

```

In [17]: # 4a. LeagueGameLog – all games in a season (team-level)
time.sleep(1)
lgl = leaguegamelog.LeagueGameLog(season=SEASON, player_or_team_abbreviation=TEAM_ABBREVIATION)
df = lgl.get_data_frames()[0]
explored['LeagueGameLog_Team'] = explore("LeagueGameLog (Team) – Every game")

time.sleep(1)

# 4b. BoxScoreTraditionalV3 – traditional per-player box score for a game
bst = boxscoretraditionalv3.BoxScoreTraditionalV3(game_id=GAME_ID)
dfs_bs = bst.get_data_frames()
explored['BoxScoreTraditionalV3_Player'] = explore("BoxScoreTraditionalV3 (Player) – Every player")
explored['BoxScoreTraditionalV3_Team'] = explore("BoxScoreTraditionalV3 (Team) – Every team")

time.sleep(1)

```

```
# 4c. BoxScoreAdvancedV3 – advanced per-player box score
bsa = boxscoreadvancedv3.BoxScoreAdvancedV3(game_id=GAME_ID)
dfs_bsa = bsa.get_data_frames()
explored['BoxScoreAdvancedV3_Player'] = explore("BoxScoreAdvancedV3 (Player)
```

LeagueGameLog (Team) – Every game's team-level result
Shape: (2460, 29) | Columns: 29

	SEASON_ID	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GAME_ID	GAME_D
0	22024	1610612738	BOS	Boston Celtics	0022400061	2024-10
1	22024	1610612750	MIN	Minnesota Timberwolves	0022400062	2024-10
2	22024	1610612747	LAL	Los Angeles Lakers	0022400062	2024-10

Data Types:

SEASON_ID	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
TEAM_NAME	str
GAME_ID	str
GAME_DATE	str
MATCHUP	str
WL	str
MIN	int64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
STL	int64
BLK	int64
TOV	int64
PF	int64
PTS	int64
PLUS_MINUS	int64
VIDEO_AVAILABLE	int64

dtype: object

```
=====
=====
📊 BoxScoreTraditionalV3 (Player) – Per-player box score
Shape: (26, 34) | Columns: 34
=====
=====
```

	gameId	teamId	teamCity	teamName	teamTricode	teamSlug	personId	first
0	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1631200	
1	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1641739	
2	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1642270	I

📋 Data Types:

gameId	str
teamId	int64
teamCity	str
teamName	str
teamTricode	str
teamSlug	str
personId	int64
firstName	str
familyName	str
nameI	str
playerSlug	str
position	str
comment	str
jerseyNum	str
minutes	str
fieldGoalsMade	int64
fieldGoalsAttempted	int64
fieldGoalsPercentage	float64
threePointersMade	int64
threePointersAttempted	int64
threePointersPercentage	float64
freeThrowsMade	int64
freeThrowsAttempted	int64
freeThrowsPercentage	float64
reboundsOffensive	int64
reboundsDefensive	int64
reboundsTotal	int64
assists	int64
steals	int64
blocks	int64
turnovers	int64
foulsPersonal	int64
points	int64
plusMinusPoints	float64
dtype:	object

BoxScoreTraditionalV3 (Team) – Team totals for game
Shape: (4, 26) | Columns: 26

	gameId	teamId	teamCity	teamName	teamTricode	teamSlug	minutes	fieldGoalsMade	fieldGoalsAttempted	fieldGoalsPercentage	threePointersMade	threePointersAttempted	threePointersPercentage	freeThrowsMade	freeThrowsAttempted	freeThrowsPercentage	reboundsOffensive	reboundsDefensive	reboundsTotal	assists	steals	blocks	turnovers	foulsPersonal	points	startersBench
0	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	141:33	28	52	54	12	25	33	10	20	50	10	18	28	22	1	2	15	18	108	10
1	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	141:33	28	52	54	12	25	33	10	20	50	10	18	28	22	1	2	15	18	108	10
2	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	165:03	32	58	57	15	30	36	12	22	55	12	20	32	25	2	3	18	22	120	12

Data Types:

gameId str
teamId int64
teamCity str
teamName str
teamTricode str
teamSlug str
minutes str
fieldGoalsMade int64
fieldGoalsAttempted int64
fieldGoalsPercentage float64
threePointersMade int64
threePointersAttempted int64
threePointersPercentage float64
freeThrowsMade int64
freeThrowsAttempted int64
freeThrowsPercentage float64
reboundsOffensive int64
reboundsDefensive int64
reboundsTotal int64
assists int64
steals int64
blocks int64
turnovers int64
foulsPersonal int64
points int64
startersBench str
dtype: object

BoxScoreAdvancedV3 (Player) – OFF/DEF RTG, USG%, TS% per player per game
Shape: (26, 37) | Columns: 37

	gameId	teamId	teamCity	teamName	teamTricode	teamSlug	personId	fir
0	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	1642261	
1	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	202693	
2	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	203458	

Data Types:

```

gameId                str
teamId                int64
teamCity              str
teamName              str
teamTricode           str
teamSlug              str
personId              int64
firstName              str
familyName             str
nameI                 str
playerSlug            str
position              str
comment               str
jerseyNum              str
minutes               str
estimatedOffensiveRating float64
offensiveRating        float64
estimatedDefensiveRating float64
defensiveRating        float64
estimatedNetRating     float64
netRating              float64
assistPercentage       float64
assistToTurnover       float64
assistRatio            float64
offensiveReboundPercentage float64
defensiveReboundPercentage float64
reboundPercentage      float64
turnoverRatio          float64
effectiveFieldGoalPercentage float64
trueShootingPercentage float64
usagePercentage        float64
estimatedUsagePercentage float64
estimatedPace          float64
pace                  float64
pacePer40              float64
possessions            float64
PIE                    float64
dtype: object

```

```

In [19]: # 4d. BoxScoreHustleV2 – per-player hustle stats for a single game
time.sleep(1)
bsh = boxscorehustlev2.BoxScoreHustleV2(game_id=GAME_ID)
df = bsh.get_data_frames()[0]
explored['BoxScoreHustleV2'] = explore("BoxScoreHustleV2 – Hustle stats per

```

```
time.sleep(1)

# 4e. BoxScoreMatchupsV3 – defensive matchups for a game
bsm = boxscorematchupsv3.BoxScoreMatchupsV3(game_id=GAME_ID)
df = bsm.get_data_frames()[0]
explored['BoxScoreMatchupsV3'] = explore("BoxScoreMatchupsV3 – Who guarded w

time.sleep(1)

# 4f. BoxScoreSummaryV2 – game summary info (line score, officials, etc.)
bss = boxscoresummaryv2.BoxScoreSummaryV2(game_id=GAME_ID)
dfs_bss = bss.get_data_frames()
explored['BoxScoreSummary_GameSummary'] = explore("BoxScoreSummaryV2 – Game
explored['BoxScoreSummary_LineScore'] = explore("BoxScoreSummaryV2 – Line Sc

time.sleep(1)

# 4g. WinProbabilityPBP – play-by-play with win probability
# Use an older completed game since newest games may not have data
try:
    wpp = winprobabilitypbp.WinProbabilityPBP(game_id='0022400062')
    df = wpp.get_data_frames()[0]
    explored['WinProbabilityPBP'] = explore("WinProbabilityPBP – Play-by-pla
except Exception as e:
    print(f"⚠ WinProbabilityPBP failed: {e}")
    print("Columns expected: GAME_ID, EVENT_NUM, HOME_PCT, VISITOR_PCT, HOME
```

=====

=====
 BoxScoreHustleV2 – Hustle stats per player per game
Shape: (17, 31) | Columns: 31

=====

=====

	gameId	teamId	teamCity	teamName	teamTricode	teamSlug	personId	fir
0	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	1642261	
1	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	202693	
2	0022401199	1610612747	Los Angeles	Lakers	LAL	lakers	203458	

 Data Types:

```
gameId          str
teamId          int64
teamCity        str
teamName        str
teamTricode     str
teamSlug        str
personId        int64
firstName       str
familyName      str
nameI           str
playerSlug      str
position        str
comment         str
jerseyNum       str
minutes         str
points          int64
contestedShots  int64
contestedShots2pt int64
contestedShots3pt int64
deflections     int64
chargesDrawn    int64
screenAssists   int64
screenAssistPoints int64
looseBallsRecoveredOffensive int64
looseBallsRecoveredDefensive int64
looseBallsRecoveredTotal int64
offensiveBoxOuts int64
defensiveBoxOuts int64
boxOutPlayerTeamRebounds int64
boxOutPlayerRebounds int64
boxOuts         int64
dtype: object
```

```
=====
====
🏆 BoxScoreMatchupsV3 – Who guarded whom, matchup stats
  Shape: (139, 46) | Columns: 46
=====
=====
=====
```

	gameId	teamId	teamCity	teamName	teamTricode	teamSlug	personIdOff
0	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1631200
1	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1631200
2	0022401199	1610612757	Portland	Trail Blazers	POR	blazers	1631200

📄 Data Types:

```
=====
=====
BoxScoreSummaryV2 - Game Summary (date, arena, attendance)
Shape: (1, 14) | Columns: 14
=====
=====
```

```
/var/folders/ty/9f3qjdhj5fggspmb_4ybsnm0000gn/T/ipykernel_15077/196503023.py:17: UserWarning: BoxScoreSummaryV2 has known data availability issues. Data may be missing for games on or after 4/10/2025. Users should move to BoxScoreSummaryV3 or verify data completeness for their specific use cases and implement appropriate error handling.
bss = boxscoresummaryv2.BoxScoreSummaryV2(game_id=GAME_ID)
```

	GAME_DATE_EST	GAME_SEQUENCE	GAME_ID	GAME_STATUS_ID	GAME_STATUS
0	2025-04-13T00:00:00	14	0022401199	1	3:30

Data Types:

GAME_DATE_EST	str
GAME_SEQUENCE	int64
GAME_ID	str
GAME_STATUS_ID	int64
GAME_STATUS_TEXT	str
GAMECODE	str
HOME_TEAM_ID	int64
VISITOR_TEAM_ID	int64
SEASON	str
LIVE_PERIOD	int64
LIVE_PC_TIME	str
NATL_TV_BROADCASTER_ABBREVIATION	object
LIVE_PERIOD_TIME_BCAST	object
WH_STATUS	int64

dtype: object

BoxScoreSummaryV2 – Line Score (quarter-by-quarter)

Shape: (2, 23) | Columns: 23

	GAME_DATE_EST	GAME_SEQUENCE	GAME_ID	TEAM_ID	TEAM_ABBREVIATION
0	2025-04-13T00:00:00	14	0022401199	1610612747	LAL
1	2025-04-13T00:00:00	14	0022401199	1610612757	POR

Data Types:

```

GAME_DATE_EST      str
GAME_SEQUENCE      int64
GAME_ID            str
TEAM_ID            int64
TEAM_ABBREVIATION  str
TEAM_CITY_NAME     str
TEAM_NICKNAME      str
TEAM_WINS_LOSSES   object
PTS_QTR1           object
PTS_QTR2           object
PTS_QTR3           object
PTS_QTR4           object
PTS_OT1           object
PTS_OT2           object
PTS_OT3           object
PTS_OT4           object
PTS_OT5           object
PTS_OT6           object
PTS_OT7           object
PTS_OT8           object
PTS_OT9           object
PTS_OT10          object
PTS               object

```

dtype: object

⚠ WinProbabilityPBP failed: Expecting value: line 1 column 1 (char 0)
Columns expected: GAME_ID, EVENT_NUM, HOME_PCT, VISITOR_PCT, HOME_PTS, VISITOR_PTS, etc.

5. Lineup, On/Off Court & Matchup Endpoints

These capture lineup chemistry, player on/off impact, and season-level matchup data.

```

In [20]: from nba_api.stats.endpoints import (
    commonteamroster, leaguedashlineups, leagueseasonmatchups,
    leaguelineupviz, teamdashlineups, teamplayeronoffsummary,
    shotchartdetail, playbyplayv3
)

# 5a. CommonTeamRoster – who's on the team, position, height, weight, age
time.sleep(1)
ctr = commonteamroster.CommonTeamRoster(team_id=TEAM_ID, season=SEASON)
dfs_ctr = ctr.get_data_frames()
explored['CommonTeamRoster'] = explore("CommonTeamRoster – Players, position")
if len(dfs_ctr) > 1:
    explored['CommonTeamRoster_Coaches'] = explore("CommonTeamRoster – Coach")
time.sleep(1)

# 5b. TeamPlayerOnOffDetails – team performance with/without each player
tpoo = teamplayeronoffdetails.TeamPlayerOnOffDetails(team_id=TEAM_ID, season=SEASON)
dfs_oo = tpoo.get_data_frames()
explored['TeamPlayerOnOff_On'] = explore("TeamPlayerOnOffDetails – On Court")
explored['TeamPlayerOnOff_Off'] = explore("TeamPlayerOnOffDetails – Off Court")

```

```

time.sleep(1)

# 5c. TeamPlayerOnOffSummary – simpler on/off splits
tpos = teamplayeronoffsummary.TeamPlayerOnOffSummary(team_id=TEAM_ID, season=SEASON)
dfs_os = tpos.get_data_frames()
explored['TeamPlayerOnOffSummary_On'] = explore("TeamPlayerOnOffSummary – On")
explored['TeamPlayerOnOffSummary_Off'] = explore("TeamPlayerOnOffSummary – Off")

time.sleep(1)

# 5d. LeagueDashLineups – 5-man lineup stats
ldl = leaguedashlineups.LeagueDashLineups(season=SEASON, group_quantity=5)
df = ldl.get_data_frames()[0]
explored['LeagueDashLineups_5Man'] = explore("LeagueDashLineups (5-Man) – Five Man Lineup")

time.sleep(1)

# 5e. TeamDashLineups – team-specific lineup combos
tdl = teamdashlineups.TeamDashLineups(team_id=TEAM_ID, season=SEASON, group_quantity=5)
df = tdl.get_data_frames()[0]
explored['TeamDashLineups_5Man'] = explore("TeamDashLineups (5-Man) – Team-Specific Lineup")

time.sleep(1)

# 5f. ShotChartDetail – shot-level coordinates for a player
scd = shotchartdetail.ShotChartDetail(
    team_id=TEAM_ID, player_id=PLAYER_ID, season_nullable=SEASON,
    context_measure_simple='FGA'
)
dfs_sc = scd.get_data_frames()
explored['ShotChartDetail'] = explore("ShotChartDetail – Shot x/y coordinates")
if len(dfs_sc) > 1:
    explored['ShotChartDetail_LeagueAvg'] = explore("ShotChartDetail – League Average")

time.sleep(1)

# 5g. PlayByPlayV3 – play-by-play event log
try:
    pbp = playbyplayv3.PlayByPlayV3(game_id=GAME_ID)
    df = pbp.get_data_frames()[0]
    explored['PlayByPlayV3'] = explore("PlayByPlayV3 – Event-level play-by-play")
except Exception as e:
    print(f"⚠️ PlayByPlayV3 error: {e}")
    explored['PlayByPlayV3'] = {'note': 'endpoint failed', 'expected_cols':
        'gameId', 'actionNumber', 'clock', 'period', 'teamId', 'teamTricode',
        'personId', 'playerNameI', 'xLegacy', 'yLegacy', 'shotDistance',
        'shotResult', 'isFieldGoal', 'scoreHome', 'scoreAway', 'pointsTotal',
        'actionType', 'subType', 'descriptor', 'qualifiers'
    }

time.sleep(1)

# 5h. LeagueSeasonMatchups – player vs player matchup data
try:
    lsm = leagueseasonmatchups.LeagueSeasonMatchups(season=SEASON)
    df = lsm.get_data_frames()[0]

```

```
explored['LeagueSeasonMatchups'] = explore("LeagueSeasonMatchups - Playe
except Exception as e:
    print(f"⚠️ LeagueSeasonMatchups error: {e}")
    explored['LeagueSeasonMatchups'] = {'note': 'endpoint failed', 'expected
        'SEASON_ID', 'OFF_PLAYER_ID', 'OFF_PLAYER_NAME', 'DEF_PLAYER_ID',
        'DEF_PLAYER_NAME', 'GP', 'MATCHUP_MIN', 'PARTIAL_POSS',
        'PLAYER_PTS', 'TEAM_PTS', 'MATCHUP_AST', 'MATCHUP_TOV',
        'MATCHUP_FGM', 'MATCHUP_FGA', 'MATCHUP_FG_PCT',
        'MATCHUP_FG3M', 'MATCHUP_FG3A', 'MATCHUP_FG3_PCT'
    ]}

print("\n✅ Section 5 complete!")
```

=====

=====
📊 CommonTeamRoster – Players, position, height, weight, age
Shape: (17, 16) | Columns: 16
=====

=====

	TeamID	SEASON	LeagueID	PLAYER	NICKNAME	PLAYER_SLUG	NUM	POSITI
0	1610612747	2024	00	Jarred Vanderbilt	Jarred	jarred-vanderbilt	2	
1	1610612747	2024	00	Dalton Knecht	Dalton	dalton-knecht	4	
2	1610612747	2024	00	Gabe Vincent	Gabe	gabe-vincent	7	

📄 Data Types:

TeamID	int64
SEASON	str
LeagueID	str
PLAYER	str
NICKNAME	str
PLAYER_SLUG	str
NUM	str
POSITION	str
HEIGHT	str
WEIGHT	str
BIRTH_DATE	str
AGE	float64
EXP	str
SCHOOL	str
PLAYER_ID	int64
HOW_ACQUIRED	str

dtype: object

=====

=====
📊 CommonTeamRoster – Coaches
Shape: (8, 10) | Columns: 10
=====

=====

	TEAM_ID	SEASON	COACH_ID	FIRST_NAME	LAST_NAME	COACH_NAME	IS_ASS
0	1610612747	2024	200755	JJ	Redick	JJ Redick	
1	1610612747	2024	1778	Nate	McMillan	Nate McMillan	
2	1610612747	2024	2623	Scott	Brooks	Scott Brooks	

 Data Types:

TEAM_ID int64
SEASON str
COACH_ID int64
FIRST_NAME str
LAST_NAME str
COACH_NAME str
IS_ASSISTANT int64
COACH_TYPE str
SORT_SEQUENCE object
SUB_SORT_SEQUENCE int64
dtype: object

 TeamPlayerOnOffDetails – On Court stats per player
Shape: (1, 57) | Columns: 57

	GROUP_SET	GROUP_VALUE	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	V
0	Overall	2024-25	1610612747	LAL	Los Angeles Lakers	82	5

 Data Types:

GROUP_SET	str
GROUP_VALUE	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64

PLUS_MINUS_RANK int64
dtype: object

TeamPlayerOnOffDetails – Off Court stats per player
Shape: (24, 59) | Columns: 59

	GROUP_SET	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	VS_PLAYER_ID	VS_PI
0	On/Off Court	1610612747	LAL	Los Angeles Lakers	1631108	
1	On/Off Court	1610612747	LAL	Los Angeles Lakers	203076	
2	On/Off Court	1610612747	LAL	Los Angeles Lakers	1629029	

Data Types:

GROUP_SET	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
TEAM_NAME	str
VS_PLAYER_ID	int64
VS_PLAYER_NAME	str
COURT_STATUS	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64

PFD_RANK int64
PTS_RANK int64
PLUS_MINUS_RANK int64
dtype: object

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 TeamPlayerOnOffSummary – On Court summary
Shape: (1, 57) | Columns: 57

=====

	GROUP_SET	GROUP_VALUE	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	W
0	Overall	2024-25	1610612747	LAL	Los Angeles Lakers	82	5

 Data Types:

GROUP_SET	str
GROUP_VALUE	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64

PLUS_MINUS_RANKint64

dtype: object

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TeamPlayerOnOffSummary – Off Court summary

Shape: (24, 13) | Columns: 13

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	GROUP_SET	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	VS_PLAYER_ID	VS_PI
0	On/Off Court	1610612747	LAL	Los Angeles Lakers	1631108	
1	On/Off Court	1610612747	LAL	Los Angeles Lakers	203076	
2	On/Off Court	1610612747	LAL	Los Angeles Lakers	1629029	

Data Types:

GROUP_SETstr

TEAM_IDint64

TEAM_ABBREVIATIONstr

TEAM_NAMEstr

VS_PLAYER_IDint64

VS_PLAYER_NAMEstr

COURT_STATUSstr

GPint64

MINfloat64

PLUS_MINUSfloat64

OFF_RATINGfloat64

DEF_RATINGfloat64

NET_RATINGfloat64

dtype: object

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LeagueDashLineups (5-Man) – Five-man lineup NET_RATING, MIN

Shape: (2000, 58) | Columns: 58

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	GROUP_SET	GROUP_ID	GROUP_NAME	TEAM_ID	TEAM_ABBREVIATION	GP	W
0	Lineups	-1626157-	K. Towns - O. Anunoby - J. Hart - M....	1610612752	NYK	48	31
		1628384-					
		1628404-					
		1628969-					
1	Lineups	162...	M. Conley - R. Gobert - J. Randle - ...	1610612750	MIN	55	36
		-201144-					
		203497-					
		203944-					
2	Lineups	1630162-	T. Harris - T. Hardaway Jr. - C. Cun...	1610612765	DET	39	24
		1630183-					
		-202699-					
		203501-					
		1630595-					
		1631105-					
		1641709-					

 Data Types:

GROUP_SET	str
GROUP_ID	str
GROUP_NAME	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64

PLUS_MINUS_RANK int64
SUM_TIME_PLAYED int64
dtype: object

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=====
 TeamDashLineups (5-Man) – Team-specific five-man lineups
Shape: (1, 57) | Columns: 57
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	GROUP_SET	GROUP_VALUE	TEAM_ID	TEAM_ABBREVIATION	TEAM_NAME	GP	\
0	Overall	2024-25	1610612747	LAL	Los Angeles Lakers	82	5

 Data Types:

GROUP_SET	str
GROUP_VALUE	str
TEAM_ID	int64
TEAM_ABBREVIATION	str
TEAM_NAME	str
GP	int64
W	int64
L	int64
W_PCT	float64
MIN	float64
FGM	int64
FGA	int64
FG_PCT	float64
FG3M	int64
FG3A	int64
FG3_PCT	float64
FTM	int64
FTA	int64
FT_PCT	float64
OREB	int64
DREB	int64
REB	int64
AST	int64
TOV	float64
STL	int64
BLK	int64
BLKA	int64
PF	int64
PFD	int64
PTS	int64
PLUS_MINUS	float64
GP_RANK	int64
W_RANK	int64
L_RANK	int64
W_PCT_RANK	int64
MIN_RANK	int64
FGM_RANK	int64
FGA_RANK	int64
FG_PCT_RANK	int64
FG3M_RANK	int64
FG3A_RANK	int64
FG3_PCT_RANK	int64
FTM_RANK	int64
FTA_RANK	int64
FT_PCT_RANK	int64
OREB_RANK	int64
DREB_RANK	int64
REB_RANK	int64
AST_RANK	int64
TOV_RANK	int64
STL_RANK	int64
BLK_RANK	int64
BLKA_RANK	int64
PF_RANK	int64
PFD_RANK	int64
PTS_RANK	int64

PLUS_MINUS_RANKint64
dtype: object

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 ShotChartDetail – Shot x/y coordinates, zone, distance, result
Shape: (1270, 24) | Columns: 24

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	GRID_TYPE	GAME_ID	GAME_EVENT_ID	PLAYER_ID	PLAYER_NAME	TEAM_ID
0	Shot Chart Detail	0022400015	14	2544	LeBron James	1610612747
1	Shot Chart Detail	0022400015	36	2544	LeBron James	1610612747
2	Shot Chart Detail	0022400015	145	2544	LeBron James	1610612747

 Data Types:
GRID_TYPEstr
GAME_IDstr
GAME_EVENT_IDint64
PLAYER_IDint64
PLAYER_NAMEstr
TEAM_IDint64
TEAM_NAMEstr
PERIODint64
MINUTES_REMAININGint64
SECONDS_REMAININGint64
EVENT_TYPEstr
ACTION_TYPEstr
SHOT_TYPEstr
SHOT_ZONE_BASICstr
SHOT_ZONE_AREAstr
SHOT_ZONE_RANGEstr
SHOT_DISTANCEint64
LOC_Xint64
LOC_Yint64
SHOT_ATTEMPTED_FLAGint64
SHOT_MADE_FLAGint64
GAME_DATEstr
HTMstr
VTMstr
dtype: object

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 ShotChartDetail – League Average
Shape: (20, 7) | Columns: 7

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	GRID_TYPE	SHOT_ZONE_BASIC	SHOT_ZONE_AREA	SHOT_ZONE_RANGE	FGA	FGM
0	League Averages	Above the Break 3	Back Court(BC)	Back Court Shot	38	
1	League Averages	Above the Break 3	Center(C)	24+ ft.	17147	59
2	League Averages	Above the Break 3	Left Side Center(LC)	24+ ft.	26652	95

```
 Data Types:
GRID_TYPE          str
SHOT_ZONE_BASIC    str
SHOT_ZONE_AREA     str
SHOT_ZONE_RANGE    str
FGA                int64
FGM                int64
FG_PCT             float64
dtype: object
=====
=====
```

```
 PlayByPlayV3 – Event-level play-by-play data
Shape: (492, 24) | Columns: 24
=====
=====
```

	gameId	actionNumber	clock	period	teamId	teamTricode	personId
0	0022401199	2	PT12M00.00S	1	0		0
1	0022401199	4	PT12M00.00S	1	1610612757	POR	1642270
2	0022401199	7	PT11M45.00S	1	1610612757	POR	1629680

```
 Data Types:
```

```
gameId          str
actionNumber    int64
clock           str
period          int64
teamId          int64
teamTricode     str
personId        int64
playerName      str
playerNameI     str
xLegacy         int64
yLegacy         int64
shotDistance    int64
shotResult      str
isFieldGoal     int64
scoreHome       str
scoreAway       str
pointsTotal     int64
location        str
description      str
actionType      str
subType         str
videoAvailable  int64
shotValue       int64
actionId        int64
dtype: object
```

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 LeagueSeasonMatchups – Player-vs-player season matchup stats
Shape: (141255, 27) | Columns: 27

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	SEASON_ID	OFF_PLAYER_ID	OFF_PLAYER_NAME	DEF_PLAYER_ID	DEF_PLAYER_NAI
0	22024	203076	Anthony Davis	1627734	Domantas Sabo
1	22024	203497	Rudy Gobert	1627734	Domantas Sabo
2	22024	1628973	Jalen Brunson	1630700	Dyson Dani

 Data Types:

```

SEASON_ID          str
OFF_PLAYER_ID      int64
OFF_PLAYER_NAME    str
DEF_PLAYER_ID      int64
DEF_PLAYER_NAME    str
GP                 int64
MATCHUP_MIN        str
PARTIAL_POSS       float64
PLAYER_PTS         int64
TEAM_PTS           int64
MATCHUP_AST        int64
MATCHUP_TOV        int64
MATCHUP_BLK        int64
MATCHUP_FGM        int64
MATCHUP_FGA        int64
MATCHUP_FG_PCT     float64
MATCHUP_FG3M       int64
MATCHUP_FG3A       int64
MATCHUP_FG3_PCT    float64
HELP_BLK           int64
HELP_FGM           int64
HELP_FGA           int64
HELP_FG_PERC       int64
MATCHUP_FTM        int64
MATCHUP_FTA        int64
SFL                int64
MATCHUP_TIME_SEC   float64
dtype: object

```

✅ Section 5 complete!

```

In [22]: # Summary of all explored endpoints
print(f"{'Endpoint':<40} {'Shape':<20} {'Columns':<8}")
print("=" * 70)
for name, info in explored.items():
    if isinstance(info, dict) and 'note' in info:
        print(f"{'name':<40} {'FAILED':<20} {len(info.get('expected_cols', []))}")
    elif isinstance(info, pd.DataFrame):
        print(f"{'name':<40} {str(info.shape):<20} {info.shape[1]:<8}")
    else:
        print(f"{'name':<40} {'unknown':<20}")
print(f"\nTotal endpoints explored: {len(explored)}")

```

Endpoint	Shape	Columns
TeamGameLog	(82, 27)	27
TeamEstimatedMetrics	(30, 30)	30
LeagueDashTeamStats_Base	(30, 54)	54
LeagueDashTeamStats_Advanced	(30, 46)	46
LeagueDashTeamStats_FourFactors	(30, 28)	28
LeagueDashTeamStats_Opponent	(30, 54)	54
LeagueDashTeamClutch	(30, 54)	54
LeagueHustleStatsTeam	(30, 20)	20
LeagueStandingsV3	(30, 92)	92
LeagueDashTeamPtShot	(30, 18)	18
LeagueDashPtTeamDefend	(30, 11)	11
TeamDashboardByGeneralSplits_Overall	(1, 55)	55
TeamDashboardByGeneralSplits_ByLocation	(2, 55)	55
SynergyPlayTypes_Team_Off	(0, 22)	22
LeagueDashPtStats_SpeedDistance	(30, 14)	14
LeagueDashPtStats_Passing	(30, 17)	17
LeagueDashPtStats_Possessions	(30, 20)	20
LeagueDashPtStats_Rebounding	(30, 34)	34
PlayerGameLog	(70, 27)	27
CommonPlayerInfo	(1, 33)	33
PlayerCareerStats_RegSeason	(23, 27)	27
LeagueDashPlayerStats_Base	(569, 67)	67
LeagueDashPlayerStats_Advanced	(569, 79)	79
PlayerEstimatedMetrics	(569, 32)	32
LeagueDashPlayerBioStats	(569, 23)	23
LeagueHustleStatsPlayer	(567, 28)	28
LeagueDashPlayerClutch	(498, 68)	68
LeagueDashPtStats_Player_Passing	(569, 18)	18
LeagueDashPtStats_Player_Possessions	(569, 21)	21
LeagueDashPtStats_Player_SpeedDist	(569, 15)	15
LeagueGameLog_Team	(2460, 29)	29
BoxScoreTraditionalV3_Player	(26, 34)	34
BoxScoreTraditionalV3_Team	(4, 26)	26
BoxScoreAdvancedV3_Player	(26, 37)	37
BoxScoreHustleV2	(17, 31)	31
BoxScoreMatchupsV3	(139, 46)	46
BoxScoreSummary_GameSummary	(1, 14)	14
BoxScoreSummary_LineScore	(2, 23)	23
CommonTeamRoster	(17, 16)	16
CommonTeamRoster_Coaches	(8, 10)	10
TeamPlayerOnOff_On	(1, 57)	57
TeamPlayerOnOff_Off	(24, 59)	59
TeamPlayerOnOffSummary_On	(1, 57)	57
TeamPlayerOnOffSummary_Off	(24, 13)	13
LeagueDashLineups_5Man	(2000, 58)	58
TeamDashLineups_5Man	(1, 57)	57
ShotChartDetail	(1270, 24)	24
ShotChartDetail_LeagueAvg	(20, 7)	7
PlayByPlayV3	(492, 24)	24
LeagueSeasonMatchups	(141255, 27)	27

Total endpoints explored: 50

6. Feature Summary & Model Architecture Recommendations

Model 1 — Team vs Team "Playstyle" Model

Goal: Predict win probability when Team A faces Team B, capturing *how* each team plays.

Feature Groups

Category	Source Endpoint(s)	Key Features
Offensive Efficiency	LeagueDashTeamStats_Advanced	OFF_RATING , NET_RATING , TS_PCT , EFG_PCT , AST_PCT , AST_TO , AST_RATIO
Defensive Efficiency	LeagueDashTeamStats_Advanced , LeagueDashTeamStats_Opponent	DEF_RATING , OPP_FG_PCT , OPP_FG3_PCT , OPP_PTS_PAINT , OPP_PTS_FB , OPP_TOV
Four Factors	LeagueDashTeamStats_FourFactors	EFG_PCT , FTA_RATE , TM_TOV_PCT , OREB_PCT (+ opponent mirror)
Pace & Tempo	LeagueDashPtStats_SpeedDistance , LeagueDashPtStats_Possessions	AVG_SPEED , AVG_SPEED_OFF , AVG_SPEED_DEF , PACE , POSS , TIME_OF_POSS , TOUCHES
Shot Profile	LeagueDashTeamPtShot , ShotChartDetail	FGA_FREQUENCY by zone, FG_PCT by zone, FG2A_FREQUENCY , FG3A_FREQUENCY , shot zone distributions
Passing & Ball Movement	LeagueDashPtStats_Passing	PASSES_MADE , PASSES_RECEIVED , AST , SECONDARY_AST , POTENTIAL_AST , AST_ADJ
Rebounding	LeagueDashPtStats_Rebounding	OREB , DREB , OREB_CONTEST , OREB_UNCONTEST , DREB_CHANCES , REB_CHANCE_PCT
Defense Tracking	LeagueDashPtTeamDefend	D_FGA , D_FGM , D_FG_PCT , NORMAL_FG_PCT ,

Category	Source Endpoint(s)	Key Features
		PCT_PLUSMINUS (vs league avg by distance)
Hustle / Effort	LeagueHustleStatsTeam	CONTESTED_SHOTS , DEFLECTIONS , LOOSE_BALLS_RECOVERED , CHARGES_DRAWN , SCREEN_ASSISTS
Clutch Performance	LeagueDashTeamClutch	All base stats filtered to clutch situations (last 5 min, ≤5 pts) — W_PCT , FG_PCT , NET_RATING
Play Type Tendencies	SynergyPlayTypes	POSS_PCT and PPP for Transition, Isolation, P&R Ball Handler, P&R Roll Man, Post-Up, Spot-Up, Handoff, Cut, Off-Screen
Standings / Form	LeagueStandingsV3	W_PCT , L10 record, STREAK , ConferenceGamesWon , HOME_RECORD , ROAD_RECORD , OT_RECORD
Home / Away Splits	TeamDashboardByGeneralSplits	All stats split by Home vs Away — captures home-court advantage in playstyle
Lineup Chemistry	LeagueDashLineups_5Man , TeamDashLineups_5Man	NET_RATING , PLUS_MINUS , MIN for top lineups — weighted by minutes

Derived / Engineered Features

- **Δ-features:** Difference between Team A and Team B for every metric (e.g., OFF_RATING_A – DEF_RATING_B)
- **Pace-adjusted stats:** Per-100-possession normalization
- **Rolling form:** Last 5/10/15 game moving averages from TeamGameLog
- **Rest days:** Days since last game (from TeamGameLog dates)
- **Home/Away indicator**
- **Head-to-head history:** Prior matchup results from LeagueGameLog

Model 2 — Player Model (feeds into Team Model)

Goal: Quantify each player's individual contribution across multiple dimensions, then aggregate into team-level features.

Feature Groups

Category	Source Endpoint(s)	Key Features
Bio / Physical	LeagueDashPlayerBioStats , CommonPlayerInfo	PLAYER_HEIGHT_INCHES , PLAYER_WEIGHT , AGE , DRAFT_ROUND , DRAFT_NUMBER , POSITION , COUNTRY
Traditional Stats	LeagueDashPlayerStats_Base , PlayerGameLog	PTS , REB , AST , STL , BLK , TOV , FG_PCT , FG3_PCT , FT_PCT , MIN , GP
Advanced Metrics	LeagueDashPlayerStats_Advanced	OFF_RATING , DEF_RATING , NET_RATING , PIE , TS_PCT , EFG_PCT , USG_PCT , AST_PCT , AST_TO , PACE
Estimated Metrics	PlayerEstimatedMetrics	E_OFF_RATING , E_DEF_RATING , E_NET_RATING , E_AST_RATIO , E_OREB_PCT , E_DREB_PCT , E_TOV_PCT , E_USG_PCT , E_PACE
Hustle Stats	LeagueHustleStatsPlayer	CONTESTED_SHOTS , DEFLECTIONS , LOOSE_BALLS_RECOVERED , CHARGES_DRAWN , SCREEN_ASSISTS , BOX_OUTS
Clutch	LeagueDashPlayerClutch	All base + advanced stats in clutch situations
Passing	LeagueDashPtStats_Player_Passing	PASSES_MADE , PASSES_RECEIVED , AST , SECONDARY_AST , POTENTIAL_AST , AST_ADJ , AST_TO_PASS_PCT
Ball Handling	LeagueDashPtStats_Player_Possessions	TOUCHES , TIME_OF_POSS , AVG_SEC_PER_TOUCH , AVG_DRIB_PER_TOUCH , PTS_PER_TOUCH , ELBOW_TOUCHES , PAINT_TOUCHES , POST_TOUCHES
Movement	LeagueDashPtStats_Player_SpeedDist	AVG_SPEED , AVG_SPEED_OFF , AVG_SPEED_DEF ,

Category	Source Endpoint(s)	Key Features
		DIST_MILES , DIST_MILES_OFF , DIST_MILES_DEF
Shot Chart	ShotChartDetail	Per-zone FGA, FGM, FG% distributions — build a shot profile vector per player
Career Trajectory	PlayerCareerStats	Year-over-year trends in PTS , REB , AST , PER , TS_PCT — detect improvement/decline
On/Off Impact	TeamPlayerOnOffSummary	ON_NET_RATING , OFF_NET_RATING , delta = player's marginal impact on team

Aggregation Strategy (Player → Team)

- **Weighted average** by projected minutes for each player on the roster
- **Top-N players:** Aggregate only the 8-player rotation
- **Positional encoding:** Group by position (G/F/C) and aggregate within position
- **Roster composition vector:** Embed the full starting-5 + bench into a fixed-size vector

🧠 Model 3 (Suggested) — Lineup Synergy / Chemistry Model

Goal: Capture non-linear interactions between specific player combinations.

Source	Key Features
LeagueDashLineups_5Man	NET_RATING , PLUS_MINUS , MIN , W_PCT for each 5-man unit
TeamDashLineups_5Man	Team-specific lineup combos
TeamPlayerOnOffDetails	Per-player on/off court impact — enables pairwise interaction features
BoxScoreMatchupsV3	In-game player-vs-player matchup outcomes

Approach: Use the player embeddings from Model 2 as inputs. For a given 5-player lineup, concatenate or attention-pool the 5 player vectors and predict lineup **NET_RATING** . This captures synergy (e.g., spacing around a post player, dual ball-handler lineups).

🎯 Model 4 (Suggested) — Matchup Advantage Model

Goal: Predict positional/individual matchup advantages when two specific rosters clash.

Source	Key Features
LeagueSeasonMatchups	MATCHUP_FG_PCT , MATCHUP_FG3_PCT , PLAYER_PTS , PARTIAL_POSS , MATCHUP_TOV , MATCHUP_BLK , SFL — for every OFF_PLAYER vs DEF_PLAYER
BoxScoreMatchupsV3	Game-level matchup data
LeagueDashPtTeamDefend	Team defense by shot distance

Approach: For each position matchup (e.g., Team A's PG vs Team B's PG), use historical matchup data + player embeddings to predict a per-position advantage score. Sum across all 5 positions for an overall matchup advantage.

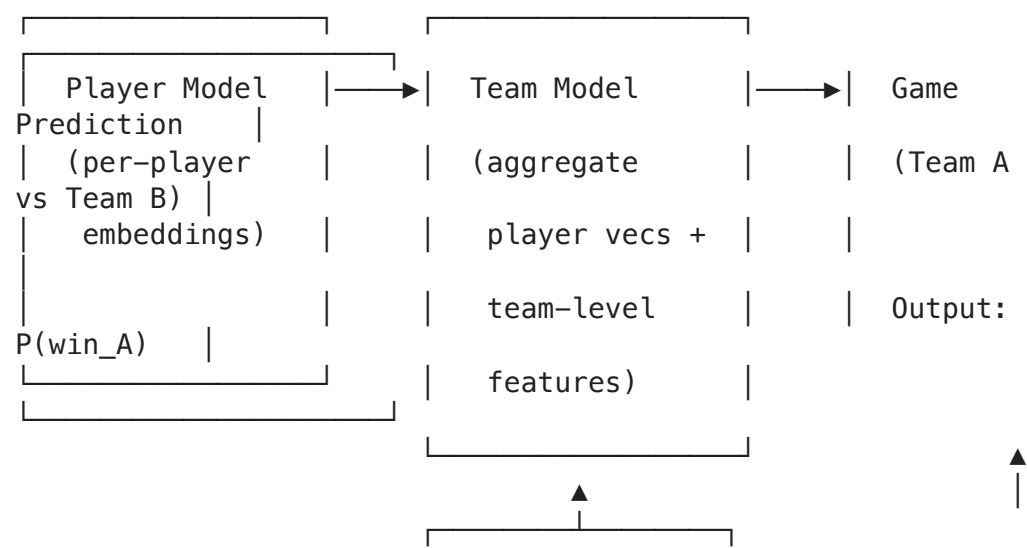
 Model 5 (Suggested) — Game State / Momentum Model

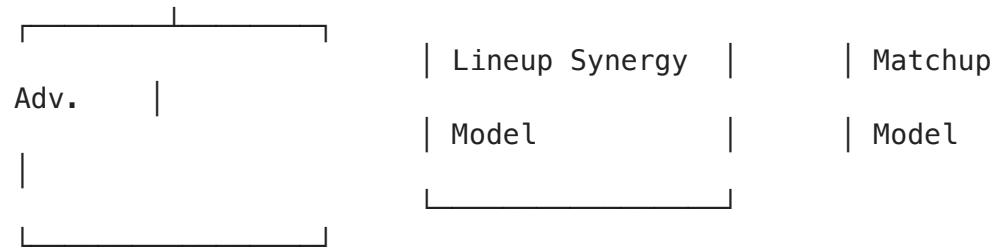
Goal: Capture in-game dynamics and momentum shifts.

Source	Key Features
PlayByPlayV3	Event sequences — runs, lead changes, timeout patterns
BoxScoreSummaryV2	Quarter-by-quarter scores (LineScore)
LeagueDashTeamClutch	Clutch-specific performance
WinProbabilityPBP	Real-time win probability curves (when available)

Approach: Sequential model (LSTM/Transformer) on play-by-play event sequences. Useful for live win-probability prediction and capturing momentum.

 Recommended Architecture Pipeline





1. **Train Player Model** → player embedding vectors (128-d or similar)
2. **Lineup Synergy Model** → lineup-level NET_RATING prediction
3. **Aggregate** player vectors + team stats → team feature vector
4. **Matchup Advantage Model** → positional edge scores
5. **Final Game Model** → takes (team_A_vec, team_B_vec, matchup_scores, home/away, rest_days, form) → **P(Team A wins)**

 Key Data Volumes (2024-25 season)

Granularity	Rows per Season	Updates
Team-level stats	30 teams	Season-end
Player-level stats	~570 players	Season-end
Game logs	~2,460 team-games (1,230 games)	Daily
Box scores	~26 player rows per game × 1,230	Daily
Play-by-play	~450 events per game × 1,230	Daily
Shot charts	~85 shots/player × 570	Daily
Matchups	~141K player-vs-player pairs	Season-end
5-man lineups	~2,000 league-wide	Season-end

```
In [ ]: # Save `explored` dict and each DataFrame in `data/` for quick reload
import os, pickle
os.makedirs('data', exist_ok=True)

saved = {'pickles': [], 'parquets': []}
try:
    with open('data/explored.pkl', 'wb') as f:
        pickle.dump(explored, f)
    saved['pickles'].append('data/explored.pkl')

    for name, obj in explored.items():
        if isinstance(obj, pd.DataFrame):
            safe = name.replace('/', '_')
            path = f'data/{safe}.parquet'
            try:
                obj.to_parquet(path, index=False)
                saved['parquets'].append(path)
            except Exception as e:
```

```
        print(f"Could not save {name} to parquet: {e}")

    print('Saved explored.pkl and parquet DataFrames to data/')
    print(saved)
except Exception as e:
    print('Error saving explored:', e)
```